Data Science & AI

DATA STRUCTURES Through Python

Linked Lists

Lecture No.- 03

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Recap of Previous Lecture

Singly Linked List Insertion

- At Beginning of SLL
- At Middle (or) at given Position
- At End of list

Topics to be Covered

SLL - Deletion

- First Node
- middle Node
- last Node

Topic : SLL - Deletion

Deletion in SLL

It can be Performed at either of 3 Positions :

1. At Beginning / First Node

2. At last / last Node

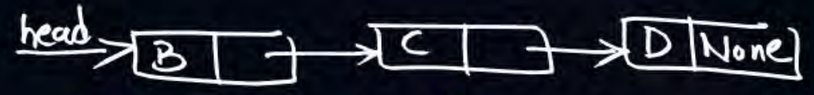
3. At given Position / middle Node

Topic : SLL - Deletion

Deletion of first Node

Without Tail / With Tail

After Deletion



Time Complexity : $O(1)$

```
def deletion_first( ):
    temp = self.head
    if self.head is None : # Empty list
        Print('Empty list')
        return
    elif self.head.next is None : # only one Element in the list
        self.head = None
        temp = None
    else :  self.head = self.head.next
            temp = None
```

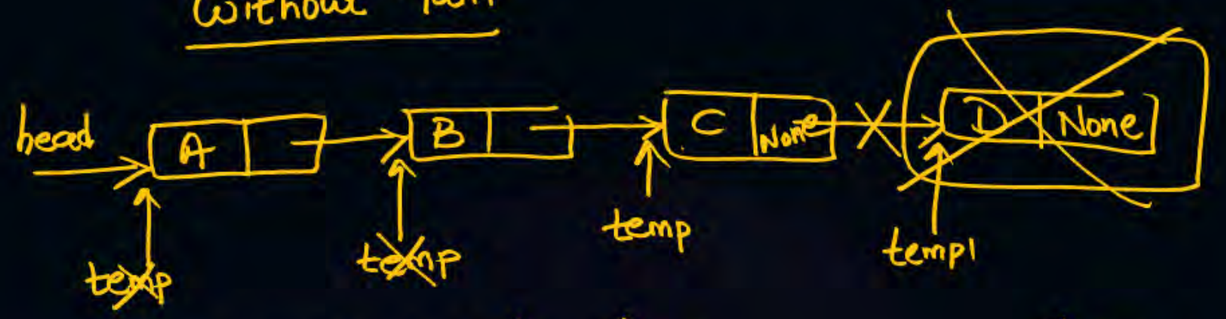
Topic : SLL - Deletion

Time Complexity : (n-1) nodes to be Traversed.

$\Rightarrow O(n-1) \Rightarrow O(n)$

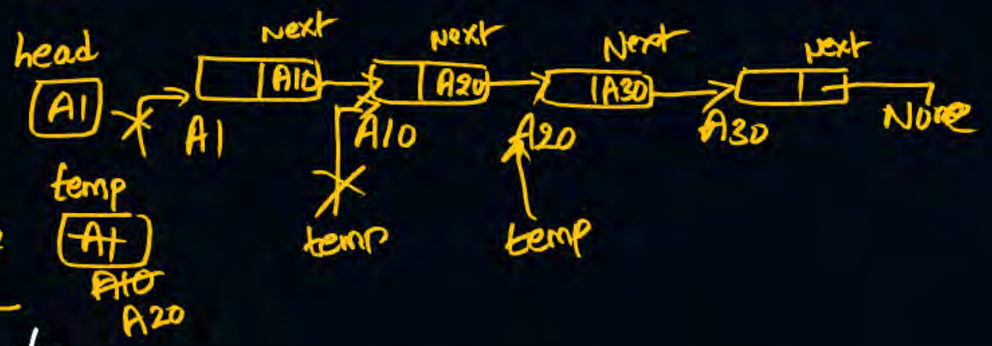
Deletion of Last Node

Without Tail



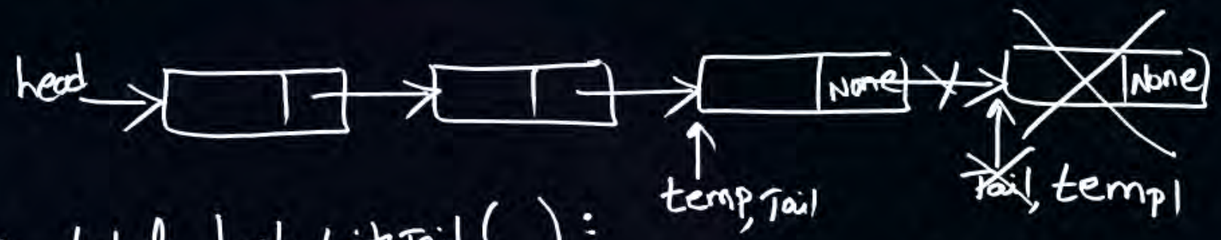

```
def delete_last( ):
    temp = self.head
    if self.head is None :
        Print('Empty list')
    elif self.head.next is None :
        self.head = None
        temp = None
    else :
        while temp.next.next is not None :
            temp = temp.next
        temp1 = temp.next
        temp.next = None
        temp1 = None
```

Topic : SLL - Deletion



Deletion of last Node with Tail at last Node

- In SLL, Traversal is Possible in one direction only.
- Though Tail at last Node, Using Tail second last Node cannot be traversed.



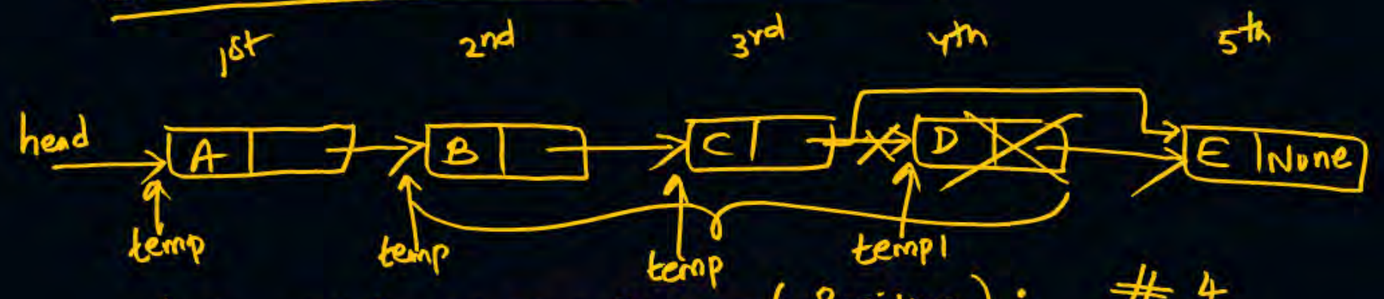
Time Complexity

$O(n)$

```
def delete_last_withTail( ):
    temp = self.head
    if self.head is None:
        Print('Empty list')
    elif self.head.next is None:
        self.head = None
        temp = None
    else:
        while temp.next.next is not None:
            temp = temp.next
        temp.next = None
        temp1 = Tail
        Tail = temp
        temp1 = None
```

Topic : SLL - Deletion

Deletion of middle Node without Tail




```
def delete_At_GivenPosition(Position):        # 4
    temp = self.head
    if self.head is None:
        Print('Empty list')
    elif self.head.next is None:
        self.head = None
            temp = None
    else:
        Count = 1
        while Count < Position-1:
            temp = temp.next
            Count = Count + 1
        temp1 = temp.next
        temp.next = temp1.next
        temp1 = None
```

Time Complexity : $O(n)$

Topic : SLL - Deletion

Deletion of middle Node with Tail : It is same as Deletion without Tail, T.C == O(n)

Deletion of a middle Node 'X' with reference of it in an object 'ob' :

Now also, we need to traverse from first Node to before Node of 'X'.

$\Rightarrow$ Time Complexity == $O(n)$

2 mins Summary

NOTE: Tail has No impact/Use in SLL in Deletion operation

	Without Tail			With Tail		
	At Front	At middle	At End	At Front	At middle	At End
Insertion	$O(1)$	$O(n)$	$O(n)$	$O(1)$	$O(n)$	$O(1)$
Deletion	$O(1)$	$O(n)$	$O(n)$	$O(1)$	$O(n)$	$O(n)$

THANK - YOU